

Table 1. SIC retrieval for the snowpack SP layer during the winter season.

True SIC (%)	Min Retrieved (%)	Max Retrieved (%)	Range (%)
Parameter: Fraction volume water			
15.0	14.3	15.0	0.7
30.0	28.7	30.0	1.3
50.0	47.8	50.0	2.2
80.0	76.5	80.0	3.5
Parameter: Radius			
15.0	14.8	15.2	0.3
30.0	29.6	30.3	0.7
50.0	49.4	50.5	1.1
80.0	79.0	80.8	1.8
Parameter: Thickness			
15.0	14.9	15.0	0.2
30.0	29.7	30.1	0.4
50.0	49.6	50.2	0.6
80.0	79.3	80.2	0.9
Parameter: Temperature			
15.0	14.8	15.1	0.2
30.0	29.7	30.1	0.4
50.0	49.5	50.2	0.7
80.0	79.2	80.4	1.2
Parameter: Density			
15.0	14.8	15.2	0.4
30.0	29.7	30.4	0.7
50.0	49.5	50.7	1.2
80.0	79.2	81.1	1.9
Parameter: Salinity			
15.0	15.0	15.0	0.0
30.0	30.0	30.0	0.0
50.0	50.0	50.0	0.0
80.0	80.0	80.0	0.0

Table 2. SIC retrieval for the depth hoar DH layer during the winter season.

True SIC (%)	Min Retrieved (%)	Max Retrieved (%)	Range (%)
Parameter: Fraction volume water			
15.0	15.0	16.0	1.0
30.0	30.0	31.9	1.9
50.0	49.9	53.2	3.2
80.0	79.9	85.1	5.2
Parameter: Radius			
15.0	14.8	15.6	0.8
30.0	29.6	31.2	1.6
50.0	49.3	52.0	2.7
80.0	79.0	83.2	4.2
Parameter: Thickness			
15.0	14.7	15.8	1.1
30.0	29.4	31.6	2.3
50.0	49.0	52.8	3.8
80.0	78.4	84.4	6.0
Parameter: Temperature			
15.0	15.0	15.1	0.1
30.0	30.0	30.2	0.2
50.0	50.0	50.3	0.4
80.0	79.9	80.6	0.6
Parameter: Density			
15.0	14.9	15.1	0.2
30.0	29.7	30.1	0.4
50.0	49.6	50.2	0.6
80.0	79.3	80.3	1.0
Parameter: Salinity			
15.0	15.0	15.0	0.0
30.0	30.0	30.0	0.0
50.0	50.0	50.0	0.0
80.0	80.0	80.0	0.0

Table 3. SIC retrieval for the sea ice SI_1 layer during the winter season.

True SIC (%)	Min Retrieved (%)	Max Retrieved (%)	Range (%)
Parameter: Thickness			
15.0	15.0	16.1	1.1
30.0	30.0	32.2	2.2
50.0	50.0	53.6	3.6
80.0	80.0	85.8	5.8
Parameter: Temperature			
15.0	14.5	15.1	0.6
30.0	29.0	30.2	1.2
50.0	48.3	50.3	2.0
80.0	77.3	80.5	3.2
Parameter: Salinity			
15.0	14.6	15.5	1.0
30.0	29.2	31.1	1.9
50.0	48.6	51.8	3.2
80.0	77.8	82.8	5.1

Table 4. Summer season: SIC retrieval for the winded snowpack SP layer.

True SIC (%)	Min Retrieved (%)	Max Retrieved (%)	Range (%)
Parameter: Fraction volume water			
15.0	14.7	16.0	1.3
30.0	29.4	32.1	2.7
50.0	49.0	53.4	4.5
80.0	78.4	85.5	7.1
Parameter: Radius			
15.0	15.0	16.2	1.3
30.0	29.9	32.5	2.5
50.0	49.9	54.1	4.2
80.0	79.8	86.6	6.8
Parameter: Thickness			
15.0	14.9	15.1	0.2
30.0	29.8	30.2	0.4
50.0	49.7	50.3	0.6
80.0	79.5	80.5	1.0
Parameter: Temperature			
15.0	15.0	15.1	0.1
30.0	30.0	30.1	0.1
50.0	50.0	50.2	0.2
80.0	79.9	80.3	0.3
Parameter: Density			
15.0	15.0	15.1	0.1
30.0	29.9	30.1	0.2
50.0	49.8	50.2	0.4
80.0	79.7	80.3	0.6
Parameter: Salinity			
15.0	15.0	15.0	0.0
30.0	30.0	30.0	0.0
50.0	50.0	50.0	0.0
80.0	80.0	80.0	0.0

Table 5. Summer season: SIC retrieval for the depth hoar DH layer.

True SIC (%)	Min Retrieved (%)	Max Retrieved (%)	Range (%)
Parameter: Fraction volume water			
15	14.95	15.87	0.92
30	29.89	31.74	1.84
50	49.82	52.90	3.07
80	79.72	84.64	4.92
Parameter: Radius			
15	14.43	15.29	0.86
30	28.86	30.58	1.72
50	48.11	50.97	2.86
80	76.97	81.55	4.58
Parameter: Thickness			
15	14.53	15.41	0.89
30	29.05	30.82	1.77
50	48.42	51.37	2.95
80	77.47	82.19	4.72
Parameter: Temperature			
15	14.99	15.10	0.11
30	29.98	30.19	0.21
50	49.96	50.32	0.35
80	79.94	80.51	0.57
Parameter: Density			
15	14.90	15.25	0.35
30	29.80	30.51	0.71
50	49.67	50.85	1.18
80	79.47	81.35	1.88
Parameter: Salinity			
15	14.99	15	0.01
30	29.98	30	0.02
50	49.96	50	0.04
80	79.94	80	0.06

Table 6. SIC retrieval for the sea ice SI_1 layer.

True SIC (%)	Min Retrieved (%)	Max Retrieved (%)	Range (%)
Parameter: Thickness			
15.0	15.0	15.8	0.8
30.0	30.0	31.5	1.6
50.0	50.0	52.6	2.6
80.0	79.9	84.1	4.2
Parameter: Temperature			
15.0	13.7	15.0	1.3
30.0	27.5	30.1	2.6
50.0	45.8	50.1	4.3
80.0	73.3	80.2	6.9
Parameter: Salinity			
15.0	14.8	15.3	0.5
30.0	29.5	30.6	1.1
50.0	49.2	51.0	1.8
80.0	78.7	81.5	2.8

Table 7. Summer season: SIC retrieval for thin ice at varying salinity levels (10.0 PSU to 14.0 PSU).

Salinity = 10.0 PSU							
True SIC (%)	0%	10%	30%	50%	80%	100%	Range
15.0	15.0	14.9	14.6	14.4	14.0	13.8	1.2
30.0	30.0	29.8	29.2	28.8	28.0	27.5	2.5
50.0	50.0	49.6	48.8	47.9	46.7	45.9	4.1
80.0	80.0	79.3	78.0	76.7	74.7	73.4	6.6
Salinity = 11.0 PSU							
15.0	15.0	14.8	14.5	14.2	13.7	13.4	1.6
30.0	30.0	29.7	29.0	28.4	27.4	26.8	3.2
50.0	50.0	49.5	48.4	47.3	45.7	44.7	5.3
80.0	80.0	79.2	77.4	75.7	73.2	71.5	8.5
Salinity = 12.0 PSU							
15.0	15.0	14.8	14.4	14.0	13.4	13.0	2.0
30.0	30.0	29.6	28.8	28.0	26.8	26.0	4.0
50.0	50.0	49.3	48.0	46.7	44.7	43.4	6.6
80.0	80.0	78.9	76.8	74.7	71.5	69.4	10.6
Salinity = 13.0 PSU							
15.0	15.0	14.8	14.3	13.8	13.1	12.6	2.4
30.0	30.0	29.5	28.6	27.6	26.1	25.2	4.8
50.0	50.0	49.2	47.6	46.0	43.5	41.9	8.1
80.0	80.0	78.7	76.1	73.5	69.7	67.1	12.9
Salinity = 14.0 PSU							
15.0	15.0	14.7	14.1	13.6	12.7	12.1	2.9
30.0	30.0	29.4	28.3	27.1	25.4	24.2	5.8
50.0	50.0	49.0	47.1	45.2	42.3	40.4	9.6
80.0	80.0	78.5	75.4	72.3	67.7	64.6	15.4

Table 8. Summer season: SIC retrieval for thin ice at salinities from 15.0 PSU to 20.0 PSU.

Salinity = 15.0 PSU							
True SIC (%)	0%	10%	30%	50%	80%	100%	Range
15	15.00	14.66	13.99	13.31	12.30	11.62	3.38
30	30.00	29.32	27.97	26.62	24.59	23.24	6.76
50	50.00	48.87	46.62	44.37	40.98	38.73	11.27
80	80.00	78.20	74.59	70.98	65.58	61.97	18.03
Salinity = 16.0 PSU							
15	15.00	14.61	13.83	13.05	11.88	11.09	3.91
30	30.00	29.22	27.66	26.09	23.75	22.19	7.81
50	50.00	48.70	46.09	43.49	39.58	36.98	13.02
80	80.00	77.92	73.75	69.58	63.34	59.17	20.83
Salinity = 17.0 PSU							
15	15.00	14.55	13.66	12.77	11.43	10.54	4.46
30	30.00	29.11	27.32	25.54	22.86	21.07	8.93
50	50.00	48.51	45.54	42.56	38.09	35.12	14.88
80	80.00	77.62	72.86	68.09	60.95	56.19	23.81
Salinity = 18.0 PSU							
15	15.00	14.50	13.49	12.48	10.97	9.96	5.04
30	30.00	28.99	26.97	24.96	21.93	19.91	10.09
50	50.00	48.32	44.96	41.60	36.55	33.19	16.81
80	80.00	77.31	71.93	66.55	58.48	53.10	26.90
Salinity = 19.0 PSU							
15	15.00	14.44	13.31	12.18	10.49	9.36	5.64
30	30.00	28.87	26.61	24.36	20.97	18.72	11.28
50	50.00	48.12	44.36	40.60	34.95	31.19	18.81
80	80.00	76.99	70.97	64.95	55.93	49.91	30.09
Salinity = 20.0 PSU							
15	15.00	14.37	13.12	11.87	10.00	8.74	6.26
30	30.00	28.75	26.25	23.74	19.99	17.49	12.51
50	50.00	47.91	43.74	39.57	33.32	29.15	20.85
80	80.00	76.66	69.99	63.32	53.31	46.64	33.36

Table 9. Thin ice in the winter season: SIC retrieval at salinities from 10.0 PSU to 14.0 PSU.

Salinity = 10.0 PSU							
True SIC (%)	0%	10%	30%	50%	80%	100%	Range
15	15.00	14.97	14.90	14.82	14.71	14.63	1.54
30	30.00	29.95	29.85	29.75	29.61	29.51	3.08
50	50.00	49.92	49.76	49.59	49.36	49.20	5.13
80	80.00	79.86	79.61	79.36	79.01	78.76	8.20
Salinity = 11.0 PSU							
15	15.00	14.95	14.84	14.73	14.57	14.46	1.89
30	30.00	29.90	29.70	29.51	29.15	28.95	3.77
50	50.00	49.84	49.44	49.04	48.39	47.99	6.28
80	80.00	79.79	79.19	78.59	77.50	76.90	10.06
Salinity = 12.0 PSU							
15	15.00	14.92	14.76	14.59	14.31	14.15	2.27
30	30.00	29.85	29.53	29.20	28.62	28.30	4.55
50	50.00	49.76	49.13	48.50	47.42	46.79	7.58
80	80.00	79.61	78.53	77.45	75.52	74.45	12.13
Salinity = 13.0 PSU							
15	15.00	14.89	14.66	14.43	14.06	13.83	2.69
30	30.00	29.78	29.33	28.88	28.12	27.67	5.39
50	50.00	49.67	48.87	48.06	46.49	45.69	8.98
80	80.00	79.34	77.73	76.12	73.18	71.57	14.36
Salinity = 14.0 PSU							
15	15.00	14.86	14.55	14.24	13.77	13.46	3.15
30	30.00	29.71	29.12	28.53	27.57	26.98	6.30
50	50.00	49.55	48.68	47.81	45.92	45.05	10.49
80	80.00	79.06	77.16	75.26	71.45	69.55	16.79

Table 10. Thin ice in the winter season: SIC retrieval at salinities from 15.0 PSU to 20.0 PSU.

Salinity = 15.0 PSU							
True SIC (%)	0%	10%	30%	50%	80%	100%	Range
15	15.00	14.82	14.45	14.07	13.44	13.07	3.63
30	30.00	29.63	28.89	28.14	26.77	26.03	7.26
50	50.00	49.45	48.23	47.01	44.58	43.36	12.10
80	80.00	78.79	76.23	73.67	68.39	65.83	19.35
Salinity = 16.0 PSU							
15	15.00	14.78	14.35	13.91	13.19	12.76	4.14
30	30.00	29.56	28.70	27.83	26.11	25.26	8.28
50	50.00	49.34	47.83	46.33	43.44	41.93	13.81
80	80.00	78.53	75.26	71.98	65.75	62.48	22.09
Salinity = 17.0 PSU							
15	15.00	14.73	14.24	13.75	12.94	12.45	4.69
30	30.00	29.48	28.48	27.49	25.65	24.65	9.38
50	50.00	49.23	47.48	45.74	42.41	40.66	15.63
80	80.00	78.26	74.48	70.70	63.89	60.11	25.01
Salinity = 18.0 PSU							
15	15.00	14.68	14.14	13.59	12.69	12.15	5.25
30	30.00	29.41	28.26	27.11	25.18	24.03	10.51
50	50.00	49.12	47.00	44.88	41.21	39.09	17.51
80	80.00	77.99	73.26	68.52	61.49	56.76	28.02
Salinity = 19.0 PSU							
15	15.00	14.62	14.02	13.42	12.43	11.83	5.84
30	30.00	29.29	27.97	26.64	24.57	23.25	11.68
50	50.00	48.90	46.02	43.13	38.98	36.10	19.47
80	80.00	77.72	71.97	66.23	57.30	51.56	31.15
Salinity = 20.0 PSU							
15	15.00	14.57	13.91	13.25	12.17	11.51	6.44
30	30.00	29.17	27.67	26.17	23.96	22.46	12.88
50	50.00	48.68	45.48	42.28	37.57	34.37	21.47
80	80.00	77.45	70.96	64.46	54.72	48.23	34.35

Table 11. Ocean component of uncertainties in sea ice concentration (SIC) retrieval. Minimum, maximum, and range of retrieved SIC (%) for each true SIC level under summer and winter conditions.

True SIC (%)	Summer			Winter		
	Min	Max	Range	Min	Max	Range
15.0	12.5	26.8	14.3	12.6	26.4	13.8
30.0	27.9	39.7	11.8	28.0	39.4	11.4
50.0	48.5	56.9	8.4	48.6	56.7	8.1
80.0	79.4	82.8	3.4	79.4	82.7	3.3